

Numerical Simulation and Analysis of the Maxwell-Boltzmann Speed Distribution

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Abstract

This paper is a technical report to the Project *Numerical Simulation and Analysis of the Maxwell-Boltzmann Speed Distribution* in which we are analysing the Maxwell-Boltzmann Speed Distribution.

The Maxwell-Boltzmann distribution provides a fundamental statistical description of the speeds of particles in a classical gas at thermal equilibrium. In this work, a numerical simulation of the Maxwell-Boltzmann speed distribution was performed using Python and NumPy. Velocity components along the three Cartesian directions were generated as independent Gaussian random variables with variances determined by the temperature of the system. The speeds of individual particles were then computed from the generated velocity vectors, and normalized histograms were constructed to approximate the corresponding probability density functions.

The simulated speed distributions were compared with the analytical Maxwell-Boltzmann distribution for a range of temperatures. Excellent agreement was observed between the numerical results and theoretical predictions, validating both the simulation methodology and the underlying statistical framework. The influence of temperature on the distribution was investigated by examining the shift in the peak position, the broadening of the distribution, and the increasing population of high-speed particles at elevated temperatures.

In addition, characteristic quantities including the most probable speed, mean speed, and root-mean-square speed were calculated and compared with their theoretical values. The project demonstrates how the Maxwell-Boltzmann distribution naturally emerges from Gaussian velocity components and illustrates the connection between microscopic particle motion and macroscopic thermodynamic properties. The study serves as an introduction to computational statistical mechanics and highlights the usefulness of numerical simulations in validating theoretical models.